DIGITAL FORENSICS

UNIT I — FUNDAMENTALS OF DIGITAL FORENSICS

Study format: Topic notes followed by related exam questions and short answers. QB labels have been removed for cleaner revision.

SYLLABUS OVERVIEW

- Introduction to Digital Forensics
- Types of Cybercrimes
- Digital Evidence and Characteristics
- Chain of Custody
- Forensic Investigation Process
- Legal and Ethical Issues
- Cyber Laws and Compliance
- NIST and ISO/IEC 27037

UNIT I — FUNDAMENTALS OF DIGITAL FORENSICS

1. Introduction to Digital Forensics

Definition

Digital Forensics is the scientific process of identifying, collecting, preserving, examining, analyzing and presenting digital evidence to investigate cybercrime.

Example: Recovering deleted files and browser history from a laptop during an investigation.

Objectives

- Collect evidence related to cybercrime and preserve its integrity.
- Determine how and when an incident occurred.
- Establish connections between a suspect and an incident.
- Present findings to a court or authorized authority.

Major branches

Branch	Focus
Computer forensics	Computers, hard disks and files
Network forensics	Network traffic and connections
Mobile forensics	Smartphones, calls, messages and app data
Cloud forensics	Cloud servers, storage and activity

Define Digital Forensics and state its main objectives.

Answer: It is the scientific process of identifying, collecting, preserving, analyzing and presenting digital evidence. Its objectives are to investigate cybercrimes, preserve evidence integrity and support legal proceedings.

Identify the major branches of Digital Forensics.

Answer: Computer, network, mobile and cloud forensics.

List the advantages of Digital Forensics.

Answer: It helps investigate cybercrimes, recover deleted evidence, establish timelines, support legal proceedings and improve cybersecurity.

Differentiate between Digital Forensics and Cybersecurity.

Answer: Digital Forensics investigates incidents and evidence; Cybersecurity protects systems through prevention, detection and safeguards.

2. Types of Cybercrimes

Cybercrime: An illegal activity involving computers, digital devices or networks as a tool, target or medium.

Type	Meaning / example
Phishing	Fake messages or websites used to steal information; e.g., a fake bank login link.
Cyberstalking	Repeated online monitoring, contact or harassment.
Cyber defamation	Online content that harms a person's reputation.
Web defacement	Unauthorized alteration of a website.
Email spoofing	Forging sender details to make an email appear to come from someone else.
Identity theft	Misusing another person's identity or personal details.
Ransomware	Malware that locks or encrypts data for ransom.
Denial of Service (DoS)	Overloading a service so it becomes unavailable.

Define cybercrime and give three examples.

Answer: Cybercrime is illegal activity involving computers or networks. Examples are phishing, ransomware and identity theft.

Differentiate between Cyber Defamation and Cyber Stalking.

Answer: Cyber defamation harms a person's reputation through damaging content. Cyberstalking involves repeated online monitoring, contact, threats or harassment.

Differentiate between Web Defacement and Email Spoofing.

Answer: Web defacement is unauthorized modification of a website. Email spoofing forges sender information to disguise an email's origin.

Classify cybercrimes with suitable examples.

Answer: Cyber-dependent crimes require digital technology to exist, such as malware attacks. Cyber-enabled crimes are traditional crimes facilitated by technology, such as online fraud.

Define phishing and describe its common types.

Answer: Phishing uses fake messages or websites to steal sensitive information. Types include email phishing, spear phishing (targeted), smishing (SMS-based) and vishing (voice-call-based).

3. Digital Evidence and Its Characteristics

Digital evidence: Information stored or transmitted digitally that can help prove or disprove a fact. Examples include emails, call logs, browser history, CCTV recordings, chat messages, system logs and files.

Characteristics

- Fragile: Can be altered or deleted easily.
- Replicable: Exact copies can be created.
- Volatile: Some data disappears when power is lost.
- Hidden: May exist in deleted files, metadata or system logs.
- Time-sensitive: Timestamps can help reconstruct events.
- Integrity: Hash values can help check whether data has changed.

Type	Meaning	Examples
Volatile evidence	May disappear when a system shuts down or changes state.	RAM, running processes, active connections
Non-volatile evidence	Generally remains stored after power is off.	Hard-disk files, stored emails, USB data

Key evidence standards

Relevance: Connected to the case. Authenticity: Genuine and what it is claimed to be. Integrity: Unchanged or changes accounted for. Reliability: Methods produce trustworthy results. Sufficiency: Enough evidence to support a conclusion.

Define digital evidence and give two examples.

Answer: Digital evidence is information in electronic form used in an investigation. Examples are email records and browser history.

Why is digital evidence fragile and easy to alter?

Answer: Actions such as opening files, changing settings or shutting down a device may alter or destroy data, so evidence must be preserved and documented carefully.

Distinguish between direct and circumstantial digital evidence.

Answer: Direct evidence directly supports a fact, such as a video showing an event. Circumstantial evidence supports an inference, such as login records suggesting account access.

Explain the importance of digital evidence.

Answer: It helps identify suspects, reconstruct events, establish timelines, determine attack methods and support legal proceedings.

4. Chain of Custody (CoC)

Definition: A chronological record of evidence collection, possession, transfer, storage and handling from initial collection to presentation in court.

Steps

- Collection: Gather evidence using proper procedures.
- Documentation: Record details, date, time and collector.
- Labeling and packaging: Assign a unique ID and secure the evidence.
- Transfer: Record each transfer between authorized persons.
- Storage and examination: Keep evidence secure and record access and examination.
- Presentation: Produce evidence and custody records when required.

Importance

It helps demonstrate integrity and authenticity, records who handled evidence and when, reduces the risk of tampering, and supports credibility in court.

Define Chain of Custody and explain its steps.

Answer: It is the documented history of evidence handling. Steps include collection, documentation, labeling, packaging, transfer recording, secure storage, examination and presentation.

Give two reasons for maintaining proper Chain of Custody.

Answer: It helps show evidence was not improperly altered or substituted and establishes a documented handling history for court.

List four core conditions for digital evidence to be admissible in court.

Answer: Relevance, authenticity, integrity and reliability are common evidence principles. Actual admissibility depends on applicable law and circumstances.

Define relevance and authenticity.

Answer: Relevance means evidence logically relates to the case. Authenticity means evidence is genuine and is what the investigator claims it to be.

How can a broken Chain of Custody affect a court case?

Answer: An unexplained gap may raise doubts about alteration or substitution, reducing credibility or affecting admissibility.

What should a Chain of Custody form contain?

Answer: Case ID, evidence ID, description, collector and signature, date/time, transfer details and signatures, storage location, and examination record.

What are the consequences of a gap in the custody log?

Answer: It creates uncertainty about who controlled the evidence and may raise doubts about integrity and authenticity.

5. Forensic Investigation Process

Sequence: Identification → Preservation → Collection/Acquisition → Examination → Analysis → Documentation → Presentation

- Identification: Locate possible evidence and relevant devices/data.
- Preservation: Protect evidence from alteration, deletion or contamination.
- Collection/Acquisition: Collect evidence or create a forensic copy.
- Examination: Extract files, logs, metadata and other artifacts.
- Analysis: Interpret evidence and reconstruct events.
- Documentation: Record procedures, tools, observations and findings.
- Presentation: Prepare a report and communicate findings.

Outline the step-by-step digital forensic investigation process.

Answer: Identify evidence, preserve it, acquire it, examine and analyze the data, document findings and present the final report.

Explain identification, preservation, analysis and presentation.

Answer: Identification finds relevant evidence; preservation protects it; analysis interprets data; presentation communicates findings through a report or testimony.

Explain the basic steps of the forensic investigation process.

Answer: Identification, preservation, acquisition, examination, analysis, documentation and presentation ensure systematic collection and reliable reporting.

6. Legal and Ethical Issues

Legal issues

- Authorization: Obtain proper legal authority or consent.
- Privacy: Handle personal information carefully.
- Jurisdiction: Different laws may apply across regions or countries.
- Admissibility: Meet applicable legal requirements.
- Documentation: Maintain accurate records of actions.

Ethical responsibilities

- Maintain confidentiality and remain objective.
- Never alter or fabricate evidence.
- Use authorized tools and methods.
- Report findings accurately, including limitations.
- Do not access data beyond the authorized scope.

Explain legal compliance in collecting digital evidence.

Answer: Evidence must be collected, handled and examined according to applicable laws, authorization requirements, privacy rules and evidence-handling procedures.

State legal challenges in Digital Forensic Investigations.

Answer: Challenges include jurisdiction, privacy, lawful access, encryption, Chain of Custody and court admissibility.

How may the defence challenge digital evidence in court?

Answer: It may question authenticity, custody gaps, collection methods, possible alteration, tool reliability or interpretation.

What privacy concerns arise when examining personal devices?

Answer: Devices may contain sensitive unrelated data. Investigators should stay within the authorized scope and protect confidential information.

When should security professionals stop handling evidence and contact law enforcement?

Answer: When a serious crime may be involved, law-enforcement authority is needed, or the matter exceeds their authorization or expertise. Preserve the scene, avoid unnecessary changes and promptly contact the appropriate authority.

7. Cyber Laws and Compliance

Provision	Broad subject
IT Act, 2000	Principal Indian law addressing electronic transactions and various cyber-related offences.
Section 43	Specified unauthorized computer acts and related compensation.
Section 66	Certain Section 43 acts done dishonestly or fraudulently.
Section 66C	Identity theft.
Section 66D	Cheating by personation using computer resources.
Section 72	Breach of confidentiality and privacy in specified circumstances.

Legal note: The question bank refers to Section 65B of the Indian Evidence Act. The Bharatiya Sakshya Adhiniyam, 2023 came into force on 1 July 2024 and now provides the current framework for electronic records. Follow your course's prescribed wording in the exam.

Mention important sections of the IT Act dealing with cybercrime.

Answer: Sections 43, 66, 66C and 66D address specified unauthorized computer acts, certain computer-related offences, identity theft and cheating by personation.

Explain Section 65B and its importance for electronic records.

Answer: Section 65B of the Indian Evidence Act, 1872 set requirements for admitting certain electronic records, including certification where applicable. The current framework is under the Bharatiya Sakshya Adhiniyam, 2023.

Explain the role of compliance audits in cybersecurity.

Answer: Audits check whether an organization follows applicable laws, policies and standards; they identify gaps and support corrective action.

8. NIST and ISO/IEC 27037

NIST: National Institute of Standards and Technology. It publishes cybersecurity, incident-response and digital-forensics guidance.

- NIST SP 800-86: Integrating forensic techniques into incident response.
- NIST SP 800-61: Computer security incident handling.
- NIST Cybersecurity Framework: Framework for managing cybersecurity risk.

ISO/IEC 27037: Guidelines for identifying, collecting, acquiring and preserving digital evidence.

- Identification: Recognize potential evidence.
- Collection: Gather devices or evidence with proper control.
- Acquisition: Capture data or create a forensic copy.
- Preservation: Protect evidence from alteration or loss.

Explain the importance of the NIST Cybersecurity Framework.

Answer: It helps organizations identify risks, protect systems, detect incidents, respond to attacks and recover operations.

Explain the purpose of ISO/IEC 27037.

Answer: It guides identification, collection, acquisition and preservation of potential digital evidence.

Differentiate between NIST SP 800-86 and NIST SP 800-61.

Answer: SP 800-86 focuses on forensic techniques integrated into incident response; SP 800-61 focuses on computer security incident handling.

Compare NIST and ISO standards in digital forensics.

Answer: NIST publications provide technical guidance and recommended practices; ISO/IEC standards provide internationally recognized standards or guidelines.

Summarize the five functions of the NIST Cybersecurity Framework.

Answer: Identify: Understand assets and risks. Protect: Implement safeguards. Detect: Identify incidents. Respond: Contain and manage incidents. Recover: Restore services.

QUICK REVISION

Remember	Key points
Forensic process	Identify → Preserve → Acquire → Examine → Analyze → Document → Present
Evidence	Relevance, authenticity, integrity, reliability; volatile data can disappear.
Chain of Custody	Document who collected, transferred, stored and examined evidence.
Legal / ethical	Authorization, privacy, confidentiality, objectivity and accurate records.
NIST SP 800-86	Forensic techniques in incident response.
NIST SP 800-61	Incident handling.
ISO/IEC 27037	Identification, collection, acquisition, preservation.
NIST CSF	Identify, Protect, Detect, Respond, Recover.